CountCrypt: Quantum Cryptography between QCMA and PP

Eli Goldin, Tomoyuki Morimae, *Saachi Mutreja*, Takashi Yamakawa

Impagliazzo's Five Worlds

- **Algorithmica** — **P** = **NP**
- **Heuristica** — **P** ≠ **NP**, but problems in **NP** are easy on average.
- **Pessiland** — Hard on average problems in **NP**, but **OWF**s don't exist.
- **Minicrypt** — **OWF**s exist, **PKE** doesn't exist.
- **Cryptomania** — **PKE** exists.

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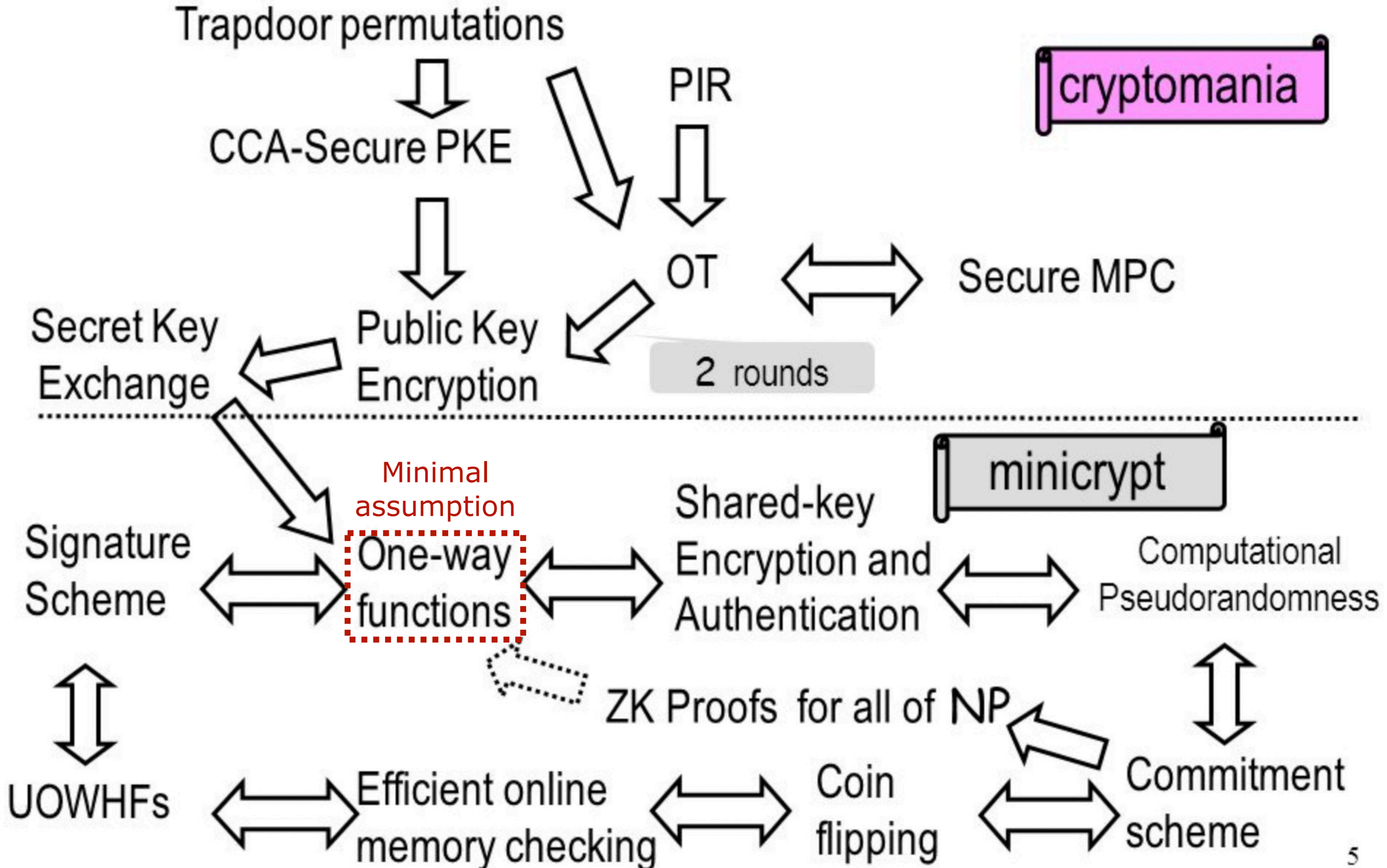
- **Minicrypt**

  OWFs exist, **PKE** doesn't exist.

Classical cryptography exists.

- **Cryptomania**

  PKE exists.


Trapdoor permutations
CCA-Secure PKE
PIR
cryptomania
OT
Secure MPC
Secret Key Exchange
Public Key Encryption
2 rounds
minicrypt
Minimal assumption
Signature Scheme
One-way functions
Shared-key Encryption and Authentication
Computational Pseudorandomness
ZK Proofs for all of NP
UOWHFs
Efficient online memory checking
Coin flipping
Commitment scheme

What about *Quantum* Cryptography?

Can quantum cryptography exist from assumptions *weaker* than one way functions?

What about *Quantum* Cryptography?

- **Algorithmica** — **P** = **NP** ⟶ Quantum cryptography possible.
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Classical cryptography exists.

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Contains inherently quantum primitives

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Qualitatively weaker than one way functions.

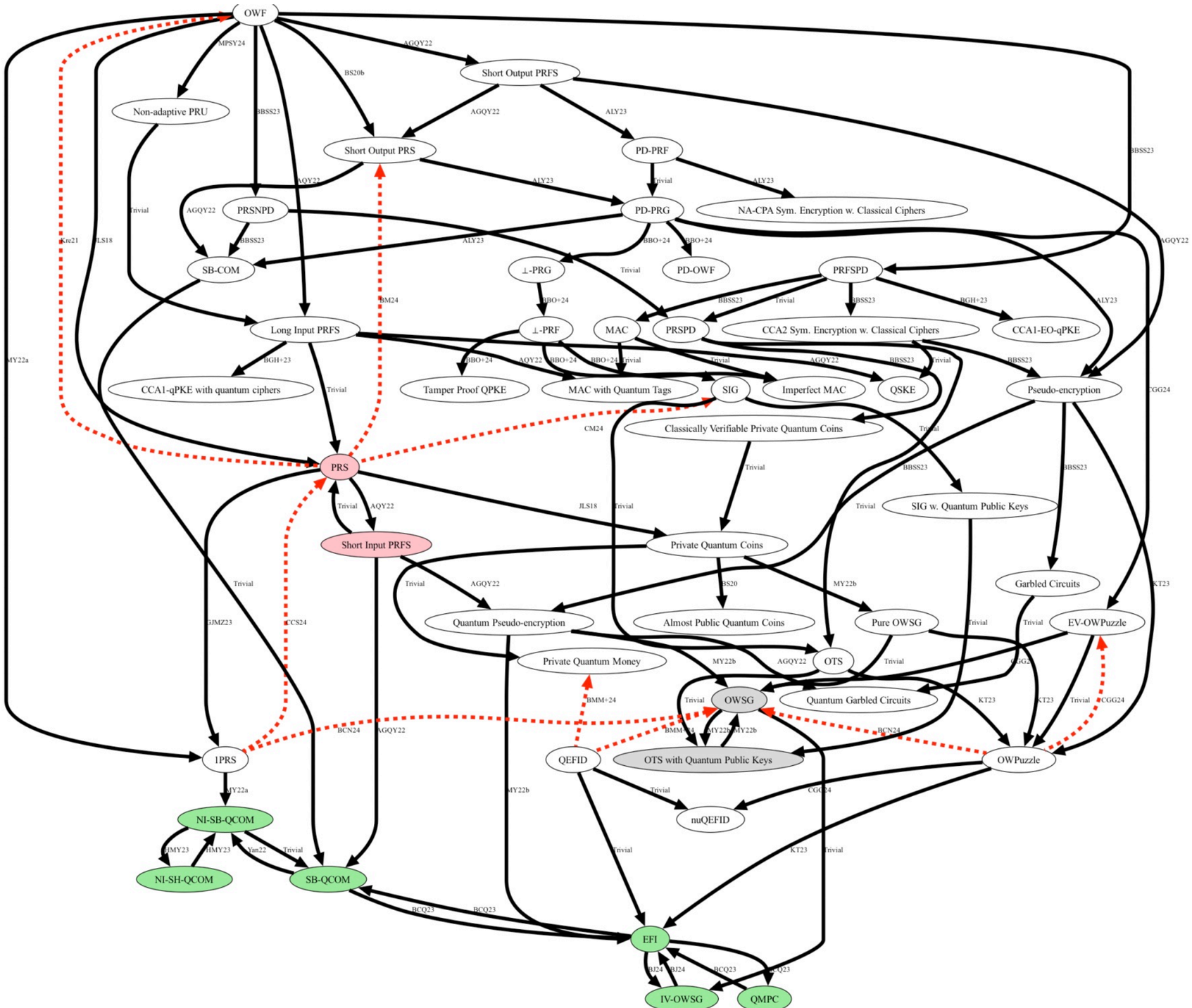
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Interesting for 2 reasons:

Qualitatively weaker than one way functions.

Useful for cryptography.


OWF
Non-adaptive PRU
Short Output PRFS
Short Output PRS
PD-PRF
PRSNPD
PD-PRG
NA-CPA Sym. Encryption w. Classical Ciphers
SB-COM
⊥-PRG
PD-OWF
PRFSPD
Long Input PRFS
⊥-PRF
MAC
PRSPD
CCA2 Sym. Encryption w. Classical Ciphers
CCA1-EO-qPKE
CCA1-qPKE with quantum ciphers
Tamper Proof QPKE
MAC with Quantum Tags
SIG
Imperfect MAC
QSKE
Pseudo-encryption
Classically Verifiable Private Quantum Coins
PRS
SIG w. Quantum Public Keys
Short Input PRFS
Private Quantum Coins
Garbled Circuits
Quantum Pseudo-encryption
Almost Public Quantum Coins
Pure OWSG
EV-OWPuzzle
Private Quantum Money
OTS
OWSG
Quantum Garbled Circuits
1PRS
QEFID
OTS with Quantum Public Keys
OWPuzzle
nuQEFID
NI-SB-QCOM
NI-SH-QCOM
SB-QCOM
EFI
IV-OWSG
QMPC
MPSY24
AGQY22
BS20b
BBSS23
ALY23
Trivial
AQY22
Kre21
JLS18
BBO+24
BGH+23
BM24
MY22a
CM24
BS20
MY22b
KT23
CGG24
GJMZ23
CCS24
BCN24
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Quantum Cryptography that breaks if

BQP=QCMA

QuantuMania

Post quantum
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Post quantum
OWFs

Efficiently verifiable
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Post quantum Cryptography

↓

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BQP=PP

CountCrypt

PRS

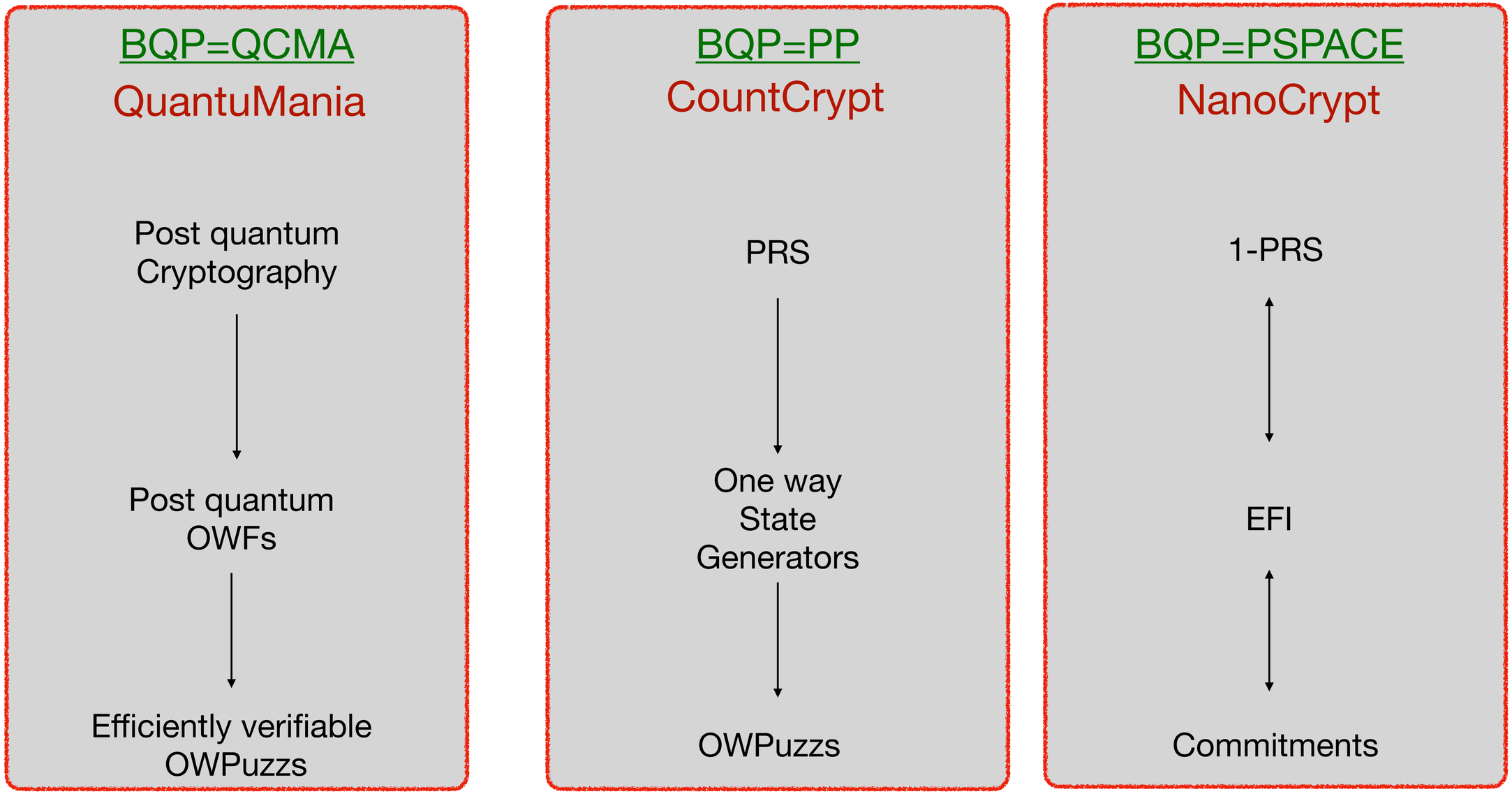
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One way State Generators

↓

OWPuzzs

Quantum Cryptography that breaks if



Is the picture complete? No 🥲

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- Many unplaced primitives:

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- Many unplaced primitives:
    - Interactive Quantum Computation Classical Communication (**QCCC**) primitives:
        - **QCCC** key exchange, **QCCC** commitments

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- Many unplaced primitives:
    - Interactive Quantum Computation Classical Communication (**QCCC**) primitives:
        - **QCCC** key exchange, **QCCC** commitments
    - Quantum Money / Quantum Lightning

Theorem: There is a unitary oracle $\mathcal{O}$ such that :

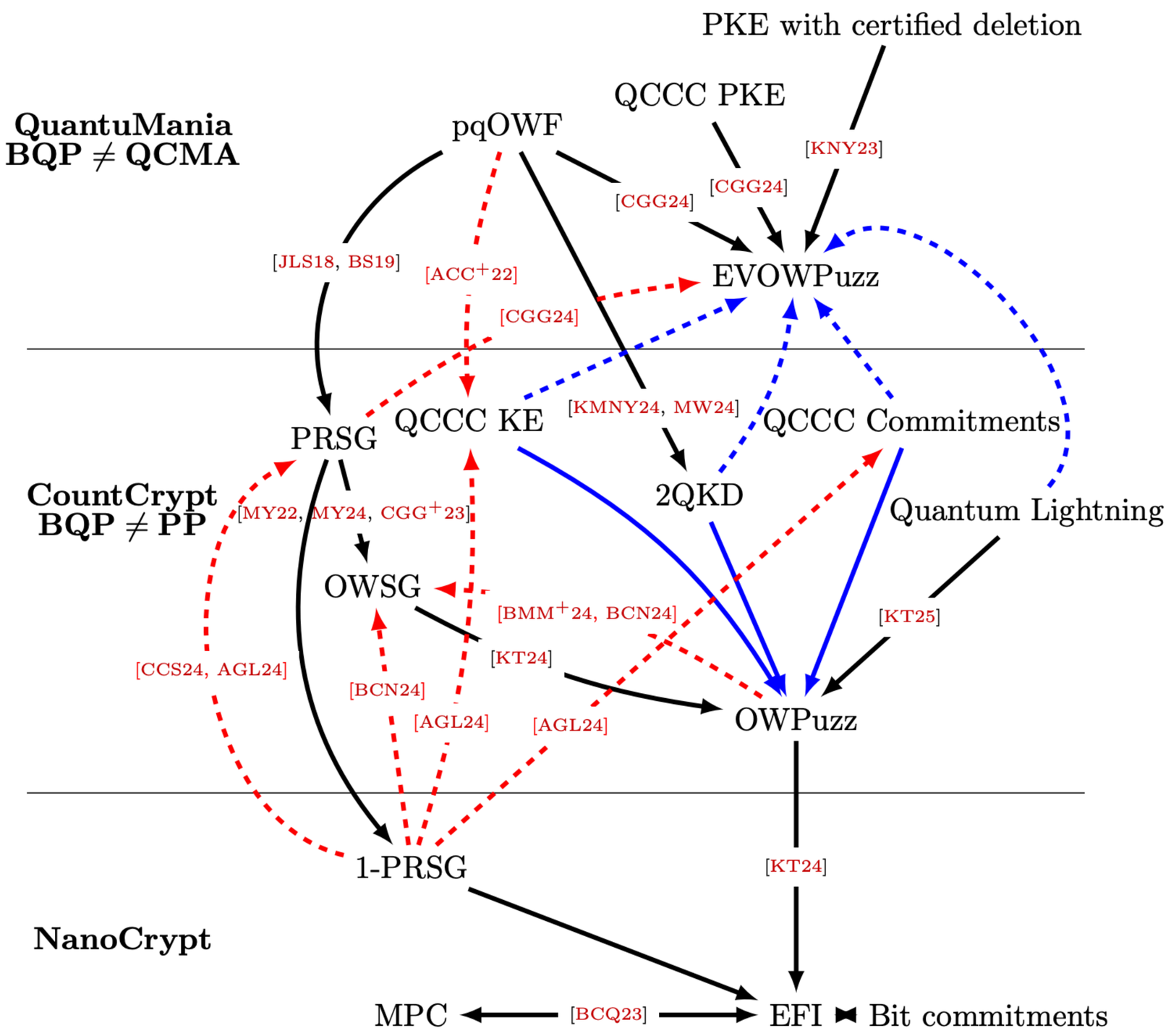
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1. $\mathbf{BQP} = \mathbf{QCMA}$.
2. Quantum Computation Classical Communication (**QCCC**) Key exchange, **QCCC** commitments, 2 round QKD exists.


PKE with certified deletion
QCCC PKE
QuantuMania
BQP ≠ QCMA
pqOWF
[KNY23]
[CGG24]
[CGG24]
[JLS18, BS19]
[ACC+22]
EVOWPuzz
[CGG24]
PRSG
QCCC KE
[KMNY24, MW24]
QCCC Commitments
CountCrypt
BQP ≠ PP
2QKD
Quantum Lightning
[MY22, MY24, CGG+23]
OWSG
[BMM+24, BCN24]
[KT25]
[KT24]
[CCS24, AGL24]
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OWPuzz
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NanoCrypt
MPC
[BCQ23]
EFI
Bit commitments

Separating Oracle $\mathcal{O}$

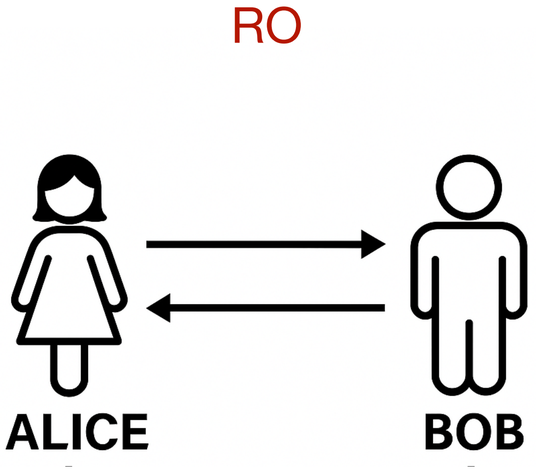
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RO: $\{0,1\}^n \times \{0,1\}^n \to \{0,1\}^n$ only accessible to Alice and Bob



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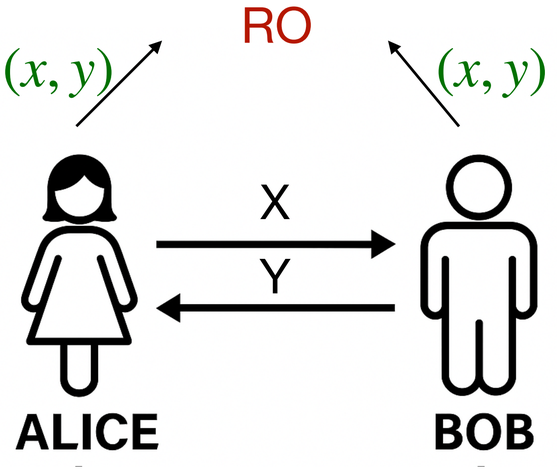
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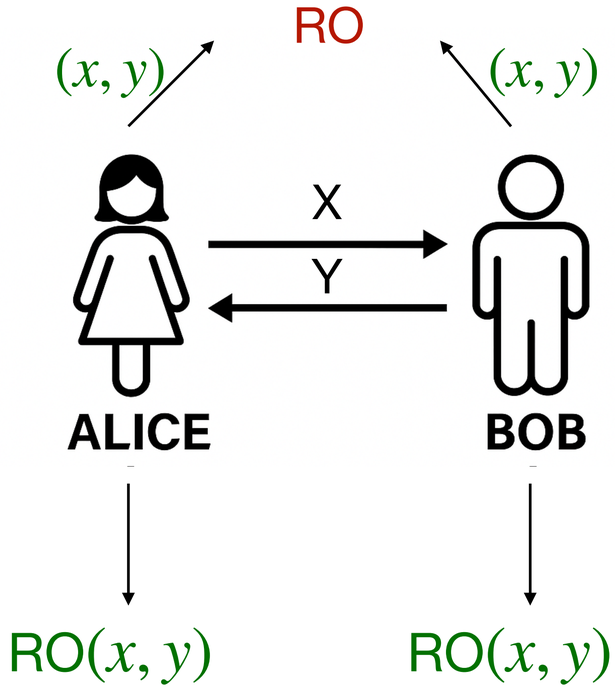
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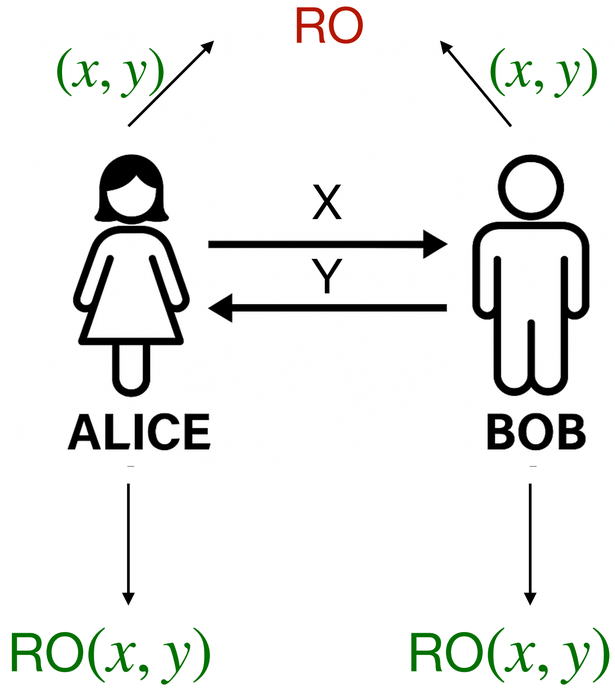


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For KE security, need RO which only allows Alice and Bob to query RO(x, y)

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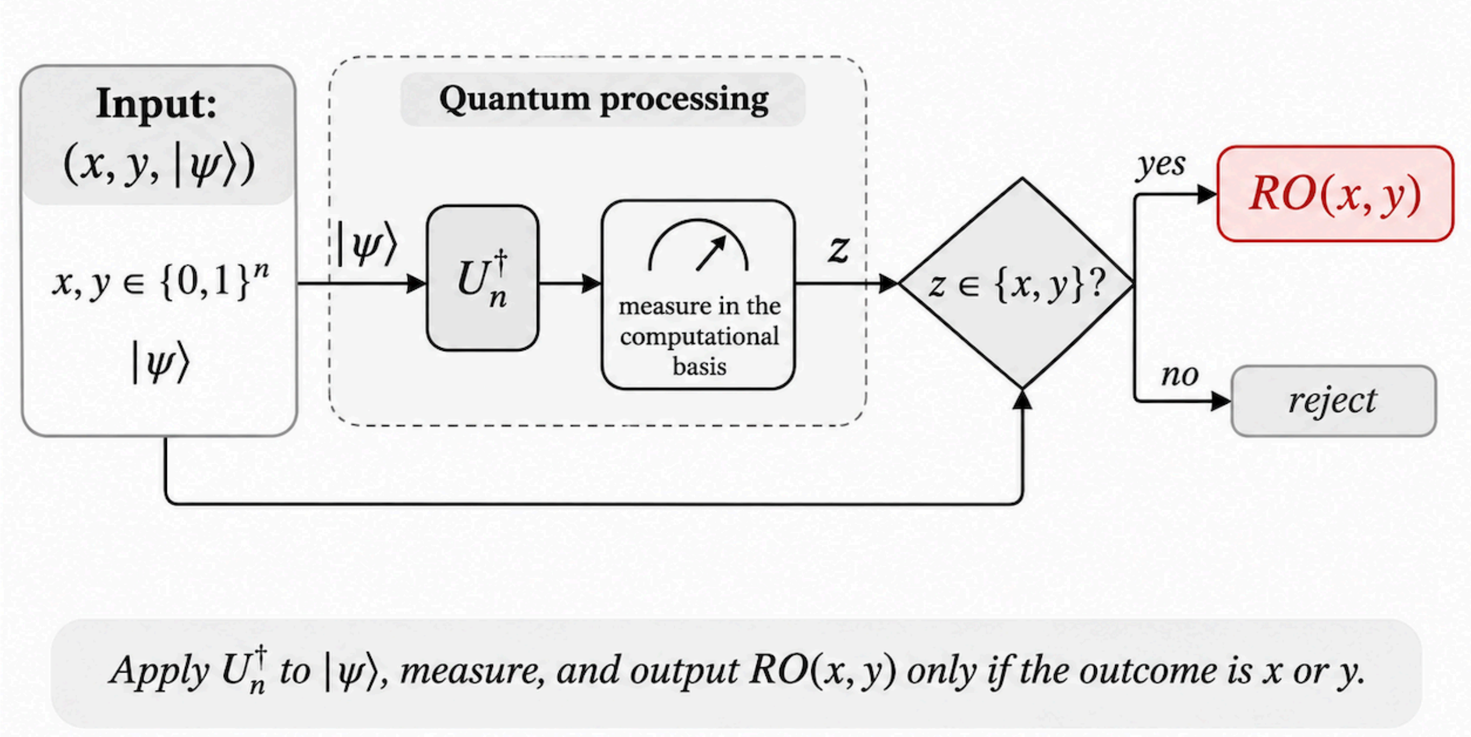
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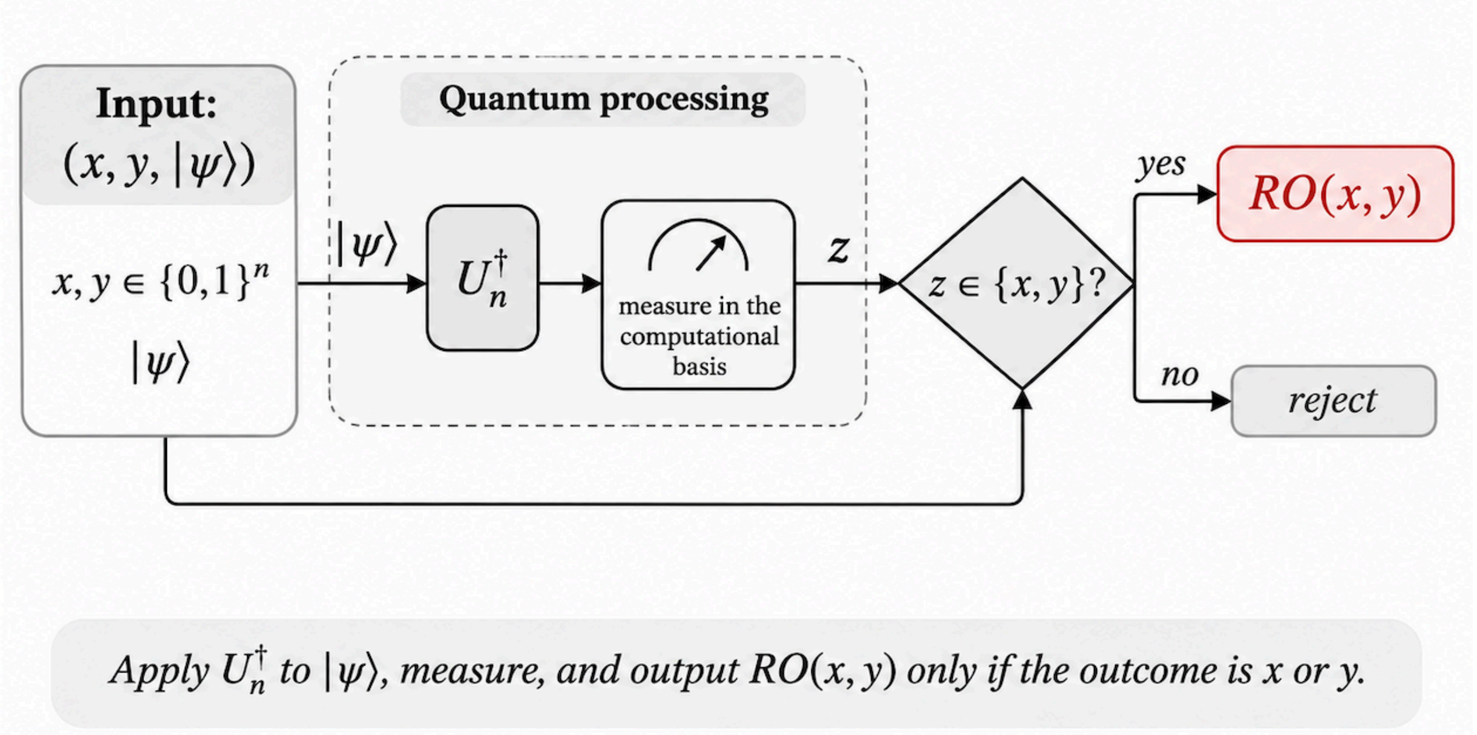


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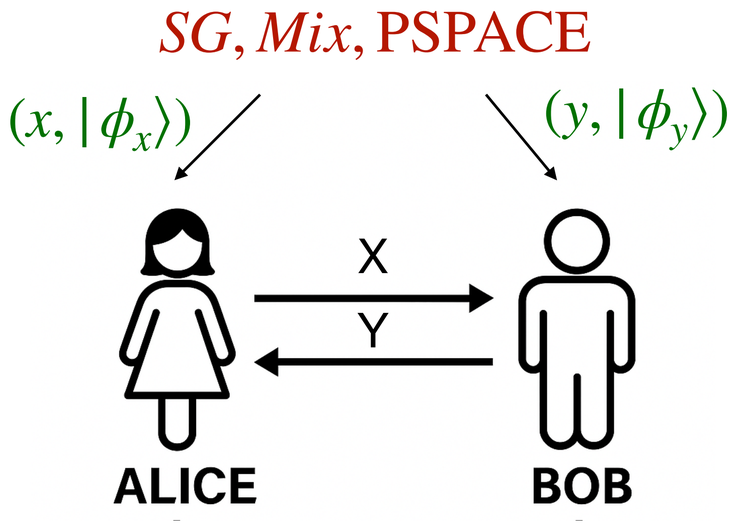


PSPACE: oracle for a PSPACE complete problem.

QCCC Key Exchange relative to $\mathcal{O}$

Alice runs $(x, |\phi_x\rangle) \leftarrow SG$ and sends x to Bob.

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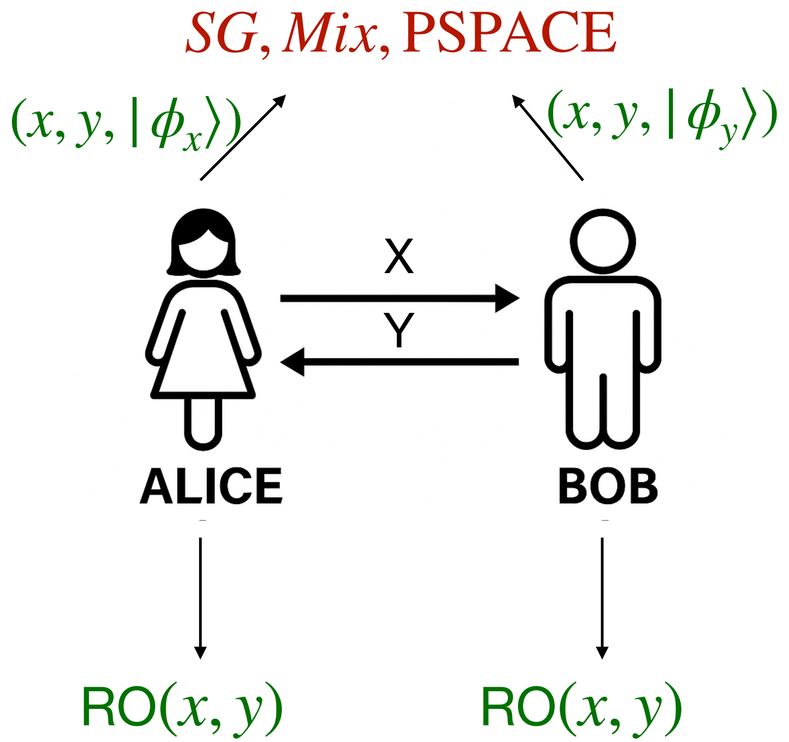
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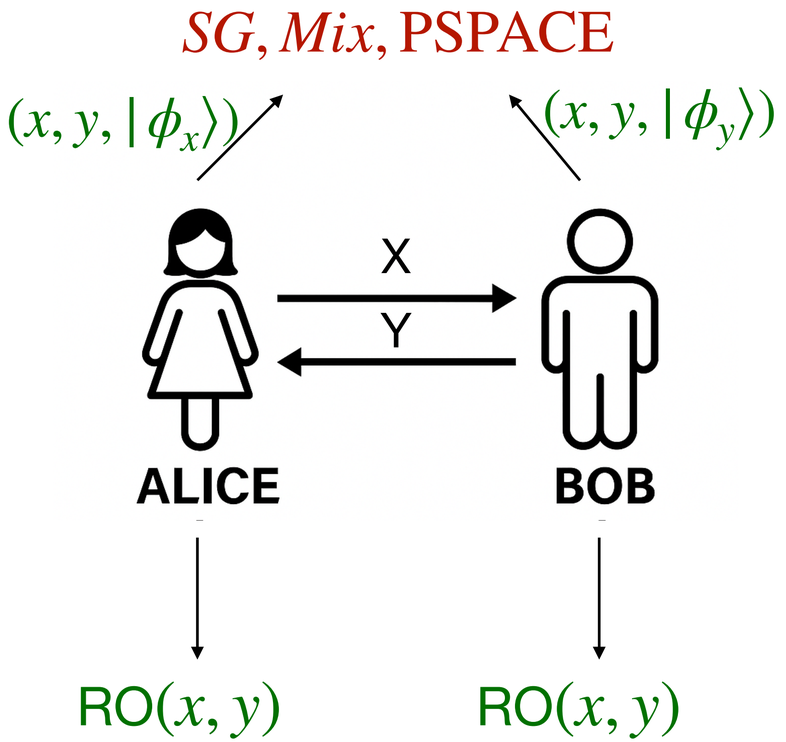
Bob outputs $\mathrm{RO}(x, y) \leftarrow Mix(x, y, |\phi_y\rangle)$.

Security Intuition

Eavesdropper cannot guess $|\phi_x\rangle, |\phi_y\rangle$ given x, y.

Impossible to query $\mathrm{RO}(x, y)$ without $|\phi_x\rangle, |\phi_y\rangle$

$\mathrm{RO}(x, y)$ appears uniformly random.



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Proof Idea [Kre21]:

QCMA algorithm cannot learn non trivial properties of unitary U, by concentration of haar measure.

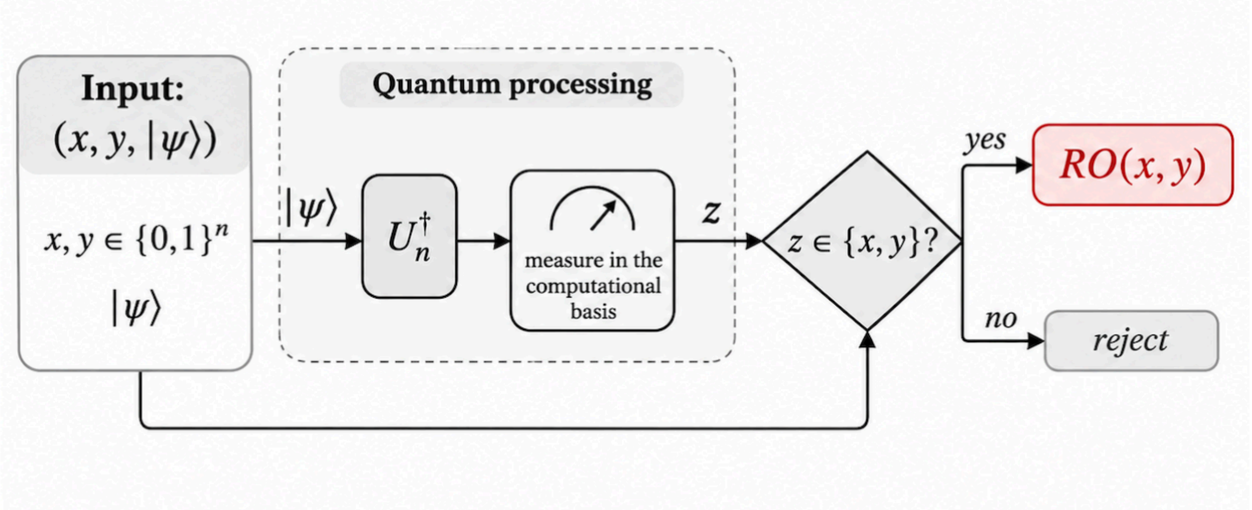
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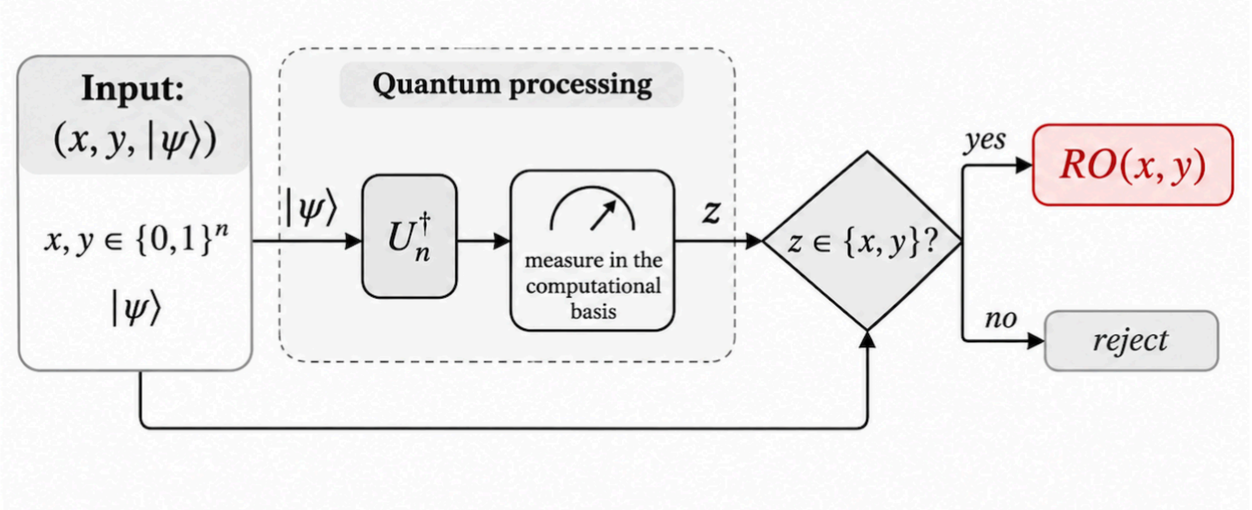
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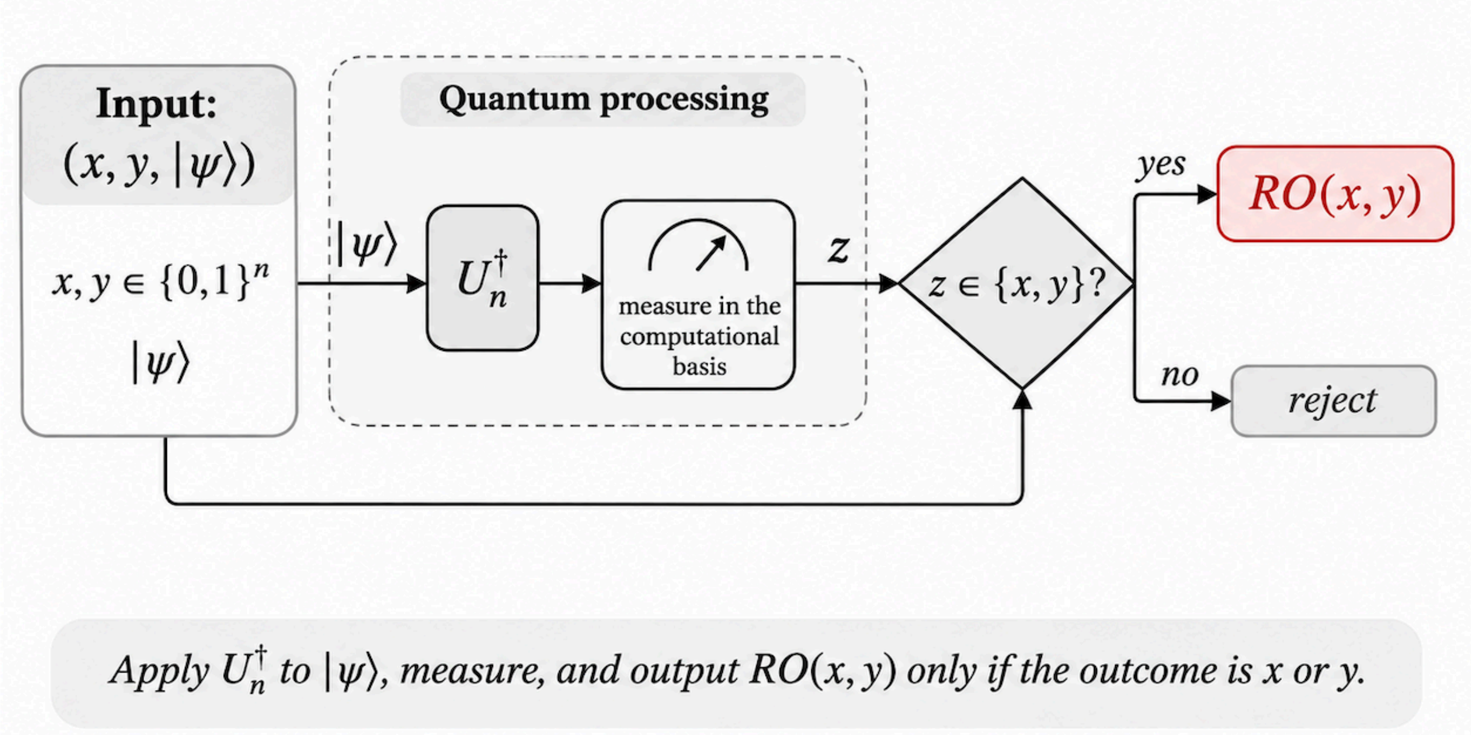
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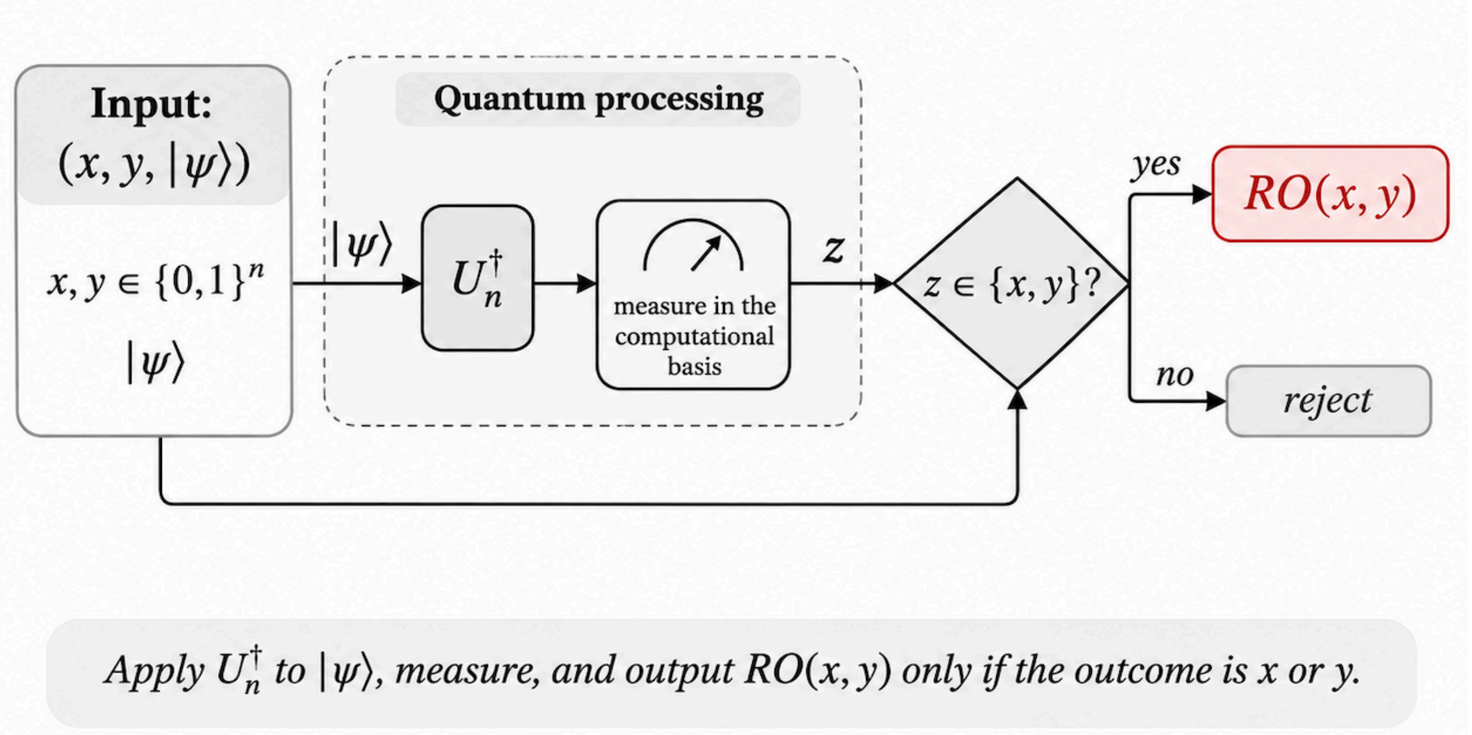
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$\mathcal{O}$ not unitary. Can't consider purified adversaries, inverse queries.

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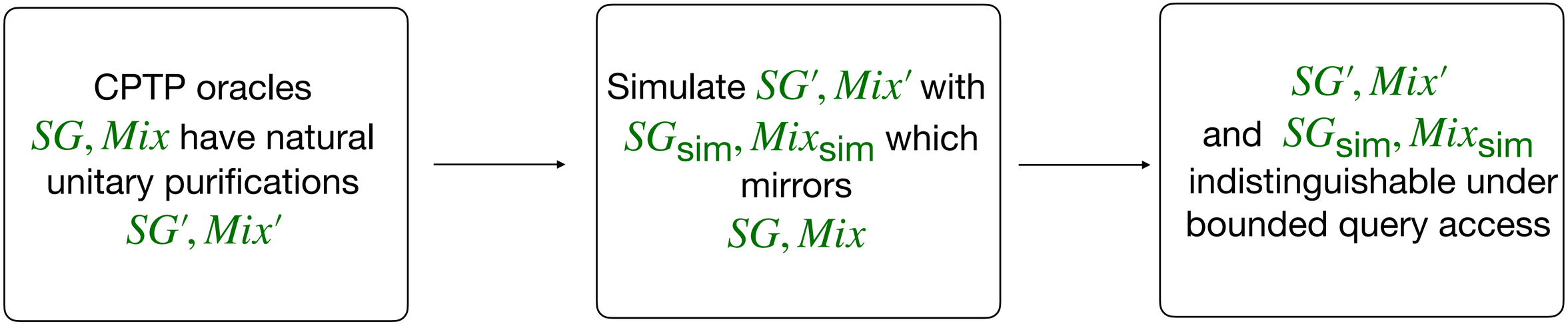
CPTP oracles
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SG', Mix' and $SG_{\text{sim}}, Mix_{\text{sim}}$ indistinguishable under bounded query access

✓

Same security intuition as SG, Mix applies.

Conclusion

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2. **QCCC** commitments, 2 round QKD exists, Quantum Lightning* exist.

Similar ideas work

Open Questions

Classical oracle $\mathcal{O}$ such that $\mathbf{BQP} = \mathbf{QCMA}$, but Quantum Lightning, **QCCC** key exchange, **QCCC** commitments exist?

Unitary oracle $\mathcal{O}$ such that $\mathbf{BQP} = \mathbf{QMA}$, but **QCCC** key exchange, **QCCC** commitments exist?

Any other separations possible with our techniques?

Thanks!